

I tried playing with the new mod and it seems to be spawning in water which is a good start but :

- 1)its jagged
- 2)it doesn't seem to behave very logically.
- 3)its very buggy
- 4)biome blending has not been fixed , the inference between biomes is still terrible and boxy.study how other mods workaround this also refer to how the default game generates biomes with a hitch.

if you look at the attached images its apparent that water is spawning which it doesnt make for it to.

rivers terminating randomly.

water flowing downstream from top of hits but there being no water uptop

water just flowing down from the side of moutains

an entire biome being covered with water

Can you study mods like:

- 1)streams reflowing
- 2)Dynamic Waters: Realistic Flowing Rivers
- 3)tectonic

all of these mods rework how rivers and waterways like underground river systems function without being buggy.

prepare a document detailing your mistakes with river system generation plus what you would infer from your study to being the proper implementation for overhauled river systems for the continent of Vantraya.

Possible solutions my suggestions:

For the biome blending issue:

To fix sharp biome borders without writing any code, you need to adjust how your biomes are placed in your world-generation settings.

Here is exactly what you need to do:

1. Overlap Your Noise Ranges

Minecraft uses numbers to decide where biomes go (like Temperature and Humidity). If Biome A stops exactly where Biome B starts, they will crash together in a sharp line.

- **The Fix:** Change the boundary numbers so they slightly overlap. For example, if Biome A ends at 0.0, make it end at 0.05. If Biome B starts at

0.0, make it start at -0.05. This forces the game engine to blend the transition space between them.

2. Group Biomes by Climate

The game cannot smoothly blend completely opposite environments.

- **The Fix:** Make sure your custom biomes are arranged logically. Never put a freezing, snowy biome directly next to a hot, dry biome. Always place a neutral "buffer" biome (like a temperate plains or a beach) between extreme climates.

3. Change Your Biome Sizes

When biomes are too small, their borders look like blocky, pixelated squares because the game's underlying noise grid is exposed.

- **The Fix:** Increase the "size" or "scale" settings of your custom biomes. Making the biomes larger gives the engine more horizontal distance to stretch and smooth out the borders.

4. Use the TerraBlender Mod

If you are using a mod-making tool (like MCreator) or just setting up a modpack, you should include a library mod called **TerraBlender**.

- **The Fix:** Adding TerraBlender handles the complicated math automatically. It tells Minecraft's world generator to treat your custom biomes exactly like vanilla biomes, forcing the game to naturally smooth out the borders for you.

Rivers issue:

1. SYNCHRONIZE CONTINENTALNESS & EROSION NOISE

----- * The Problem: Rivers try to cut through high-altitude peaks, creating massive, pixelated vertical ravines. * The Fix: Bind your river carvers exclusively to low "Continentalness" values. Ensure rivers only spawn where your terrain noise expects valleys, plains, or shorelines, keeping them entirely away from steep mountain peaks.

2. WIDEN AND SMOOTH THE RIVER CARVER SPLINES

----- * The Problem: Riverbanks form 90-degree "cookie-cutter" cliffs dropping straight down into the water. * The Fix: Open your worldgen noise settings and increase the river "Spline Width" or "Carver Radius". Instead of letting the terrain height drop instantly, adjust the threshold numbers to create a gentle, sweeping numeric slope. This forces the ground to form natural riverbanks rather than

artificial dirt walls. 3. LOWER THE NOISE OCTAVES AND FREQUENCY

----- * The Problem: The water path zig-zags in sharp, chaotic, pixelated layouts instead of curving gracefully. * The Fix: Locate the Perlin noise settings driving your river paths. Lower the "Octaves" value to 2 or 3, and significantly decrease the noise "Frequency" or "Scale". Lower frequencies stretch the noise waves out, turning jagged, glitchy streams into wide, realistic, and winding river bends. 4. ENABLE CAVER AND AQUIFER INTERPOLATION

----- * The Problem: Rivers tear open into ugly, broken, floating water pockets whenever they cross over an underground cave system. * The Fix: Check your world-generation file's carver array. Ensure that "Cave Blending" or "Aquifer Fluid Blending" is enabled with a smooth transition radius (a setting of 3 to 5 blocks). This tells the engine to ease the land geometry where a surface river intersects an underground opening.

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1. Synchronize the Continentalness and Erosion Maps

Minecraft decides where to carve rivers based on the **Erosion** noise parameter. If a river is forced to spawn in an area with a high **Continentalness** value (meaning massive, tall mountains), the depth difference is too extreme. [1]

- **The Fix:** Ensure your custom rivers are mathematically restricted to spawning *only* where your terrain's continentalness value drops into low valleys, plains, or low-lying hills. Keep rivers far away from peak mountain coordinates so they don't have to carve 80-block-deep jagged canyons.

2. Smooth Out the "River Carver" Width

If the river line cuts into the terrain like a perfect, pixelated cookie-cutter, your noise threshold is too rigid. [1]

- **The Fix:** In your noise settings, look for the **Carver** or **Spline Thickness** variables. You need to widen the river's border threshold. Instead of dropping the terrain height instantly from a value of 0.0 to -1.0, create a wide numeric slope. This forces the ground to smoothly decline into a riverbank rather than creating vertical dirt walls.

3. Lower the Noise Octaves / Frequency

If your river turns are zig-zagging in sharp, jagged 90-degree angles rather than smooth, sweeping curves, the underlying mathematical noise is firing too frequently.

- **The Fix:** Find the **frequency** or **octave** settings attached to your river noise map. Lowering the frequency makes the noise waves larger and smoother, forcing the river paths to stretch out into broad, natural bends rather than high-intensity squiggles. [\[1\]](#)

4. Enable Cave / Carver "Blending"

If your river is cutting beautifully through the surface but tearing open into chaotic, jagged holes whenever it passes over an underground cave system, the carver is overriding the terrain data incorrectly. [\[1\]](#)

- **The Fix:** Check your `noise_settings` configuration under the carver array. Ensure that **Cave Blending** is explicitly set to `true` or configured with a smooth interpolation radius. This tells the game engine to ease the geometry transition when a surface waterway intersects an underground aquifer or opening.









